ENTERPRISE DATA ARCHITECTURE & RECONCILIATION

Data Analytics Project Report

Tools Used: Python, Pandas, SQLite, SQL, Regular Expressions, Matplotlib

1. Project Objective

The objective of this project is to extract, clean, merge and analyze transaction data from different sources to calculate revenue in USD and create a financial dashboard.

2. Project Methodology

Phase 1: SQL Extraction

Used SQL Window Functions (ROW_NUMBER) to identify customers based on their latest status and extract Active customers.

Phase 2: Data Extraction

Used Pandas and Regular Expressions to extract Date, User ID, Product ID and Euro Value from server logs. Removed ERROR records.

Phase 3: Financial Reconciliation

Handled missing exchange rates using Forward Fill. Merged transaction data with exchange rates and active customer records. Calculated USD Revenue.

Phase 4: Data Analysis and Visualization

Calculated monthly revenue and identified the Top 5 Most Valuable Customers. Created two charts using Matplotlib.

3. Key Findings

- Extracted 94,902 valid transactions from server logs.
- Removed 5,098 ERROR records.
- Filled 104 missing exchange rates using Forward Fill.
- Integrated transaction, exchange rate and active customer data.
- Calculated total revenue in USD.

4. Final Results

- Total Transactions: 53,899

- Unique Active Customers: 13,888
- Missing Values: 0
- Total USD Revenue: $148,274,622.91

5. Conclusion

The project successfully reconciles transaction data, customer records and exchange rates to analyze revenue from active customers. The dashboard provides monthly revenue trends and identifies the Top 5 Most Valuable Customers.

